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MILITARY SPECIFICATION

SHOP EQUIPMENT, GENERAL-PURPOSE-REPAIR, SEMITRAILER-MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope. This specification covers a semitrailer-mounted, general-purpose-repair shop.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

AMSC N/A

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SPECIFICATIONS

FEDERAL

O-A-548 - Antifreeze/Coolant, Engine: Ethylene Glycol, Inhibited, Concentrated
W-C-375 - Circuit Breakers, Molded Case; Branch Circuit and Service
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting
VV-B-680 - Brake Fluid, Automotive
VV-F-800 - Fuel Oil, Diesel
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-1055 - Barrier Material, Waterproofed, Flexible
PPP-P-40 - Packaging and Packing of Hand Tools
PPP-T-60 - Tape: Packaging, Waterproof

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152 - Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank
MIL-T-704 - Treatment and Painting of Material
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose
MIL-C-3600 - Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of
MIL-M-3971 - Meter, Time Totalizing, Non-Hermetically Sealed, Electrical: General Specification for
MIL-E-10062 - Engines: Preparation for Shipment and Storage of
MIL-G-10924 - Grease, Automotive and Artillery
MIL-H-11511 - Heater, Space, Multifuel, With Blower, of 60,000 BTU/HR
MIL-A-11755 - Antifreeze, Arctic-Type
MIL-C-12000 - Cable, Cord, and Wire, Electric; Packaging of
MIL-C-13874 - Compressors, Reciprocating and Rotary, Air, Electric Motor Driven
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric
MIL-T-45542 - Tool Sets, Shop Sets and Kits, Maintenance, Modification and Tool, Packaging of
MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of
MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of
MIL-STD-889 - Dissimilar Metals
MIL-STD-1474 - Noise Limits for Army Materiel

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF ARMY

DRAWINGS

TA13221E9020 - Shop Equipment, General-Purpose-Repair, Semitrailer-Mounted
11021065 - Welding Requirements
11021066 - Riveting Requirements
11021251 - Finish Methods

LIST OF COMPONENTS

SC 4940-95-CL-B02 - Sets, Kits and Outfits, Components List, Shop Equipment, General Purpose Repair, Semitrailer Mounted

(Copies of specifications, standards, and other Government drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73.11 - Dimension of Plugs and Receptacles - General Purpose, 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type

ANSI C73.20 - Dimensions of Plugs and Receptacles - General Purpose, 250 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type

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ANSI C73.61 - Dimensions of Plugs and Receptacles - General Purpose, 250 Volts, 50 Amperes, 3 Phase, 3 Pole, 4 Wire, Grounding Type

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 - Standard Practice for Commercial Packaging

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1 - Motor and Generator

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, D.C. 20037.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified (see 4.3 and 6.2.1), the contractor shall furnish one complete shop for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification.

3.2 Description. The repair shop shall be in accordance with Top Assembly Drawing TA13221E9020 and as specified herein. The shop shall include all tools and equipment specified in the Set Components List as compiled from SC 4940-95-CL-B02 (see 6.3).

3.2.1 Drawings. The drawings forming a part of this specification are end-product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. The contractor is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the shop equipment.

3.3 Initial production. When specified (see 6.2.1), the contractor shall furnish to the Government one or more initial production shops for inspection.

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3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.5 Shop dimensions, weight and lifting.

3.5.1 Dimensions. Dimensions of the shop shall be as shown on the applicable drawings (see 3.2).

3.5.2 Weight. Unless otherwise specified (see 6.2.1), the gross vehicle weight with all tools and equipment shall be not more than 36,000 pounds.

3.5.3 Lifting and tiedown provisions. Lifting and tiedown provisions for the fully loaded shop are required in accordance with MIL-STD-209.

3.6 Body. The contractor shall fabricate and install the body on the chassis in accordance with the applicable drawings (see 3.2). The contractor shall furnish all items of equipment including the tools and equipment as specified in 3.2.

3.6.1 Air compressor. An electric-motor-driven air compressor shall be furnished and installed by the contractor in accordance with the applicable drawings. The air compressor shall conform to MIL-C-13374, Type I, Size 5, less air receiver tank, fittings, and inflator device.

3.6.2 Electrical power system. The contractor shall furnish and install the entire electrical power system. The system shall be module configured consisting of an alternator, control panel and related controls. The modules shall be so designed as to provide accessibility for scheduled routine and minor unscheduled maintenance without complete removal.

3.6.2.1 Alternator. The alternator shall be of the brushless type conforming to NEMA MG1, Insulation Class F. The alternator shall provide an output of 31.3 kilovolt-ampere, 25 kilowatt (Kw), 120/208 volt, three phase, 60 hertz (Hz), on a 60 percent duty cycle with a regulation of 10 percent.

3.6.2.2 Control panel. The control panel shall be of a modular design and contain all controls and instruments (time totalizing, frequency meters) necessary for the operation of the system. Controls shall be identified and functions indicated on the control panel. The control panel shall be accessible for circuit checks and component repair.

3.6.2.2.1 Frequency meter. The frequency meter shall be of the vibrating reed type with a range of 48-62 and 58-62 hertz. The meter shall have no less than five reeds at intervals not greater than one hertz. There shall be no dead zones between reeds.

3.6.2.2.2 Receptacles. Three 15 ampere, 125 volt, two pole, three wire receptacles duplex, general purpose, conforming to the ANSI C73.11 shall be furnished and installed in the control panel. The duplex receptacles shall be wired in pairs and four of the six receptacles protected with a ground fault interrupter (GFI) and located for access from ground level. Each pair of duplex receptacles shall be protected with a 15 ampere capacity circuit breaker. One 50 ampere, 250 volt, three pole, four wire receptacle, single, general purpose, three phase, conforming to ANSI C73.61 shall be furnished and installed in control panel and protected by a 50 ampere

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circuit breaker. One 15 ampere, 250 volt, two pole, three wire, single phase, general purpose receptacle, conforming to ANSI C73.20 shall be furnished and installed in control panel and protected by a 10 ampere circuit breaker. The receptacles shall be identified to indicate the voltage supplied.

3.6.2.2.3 Circuit breakers. Three 250 volt, three pole, three wire, three phase circuit breakers, of 30 ampere capacity and one each circuit breaker of 15 ampere capacity shall be provided. The circuit breakers shall conform to W-C-375, class 1A, single unit construction. Two of the 30 ampere circuit breakers shall be identified as "compressor" and "lathe".

3.6.2.2.4 Time totalizing. A time totalizing digital meter in accordance with MIL-M-3971 shall be provided to register the number of hours of engine operation.

3.6.2.3 Grounding. A 1/4-20 threaded ground stud shall be provided on the lower front of the control panel as a terminal for ground leads from the control panel mounted equipment and the alternator.

3.6.2.4 Speed controls. The vehicle engine for the alternator shall be controlled by the governor to provide 54 to 66 hertz.

3.6.2.5 Overspeed. An overspeed device shall be provided to shutdown the engine when the frequency increases to 66 Hz $\pm$1 Hz.

3.7 Electrical system. Electrical system shall be in accordance with the applicable drawings.

3.8 Tools and equipment. The tools and equipment shall be located as specified, anchored or stored to prevent displacement (see 6.8.2) during transit. Tiedown devices shall be designed for a general tool or equipment configuration to allow interchange of like items from different manufacturing sources. Whenever possible, items shall be grouped together by function.

3.8.1 Fire extinguisher. The fire extinguisher shall be mounted inside the shop and shall be accessible from the center aisle and adjacent to the rear door.

3.9 Safety. All exposed parts, assemblies or accessories, which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are hazardous to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.10 Performance.

3.10.1 Maximum power operation. When unit is operating at 208 volt, three phase and loaded to 25 Kw, there shall be no damage, deterioration, malfunction, or failure of electrical components, or failure of belts or other mechanical components as specified in 3.6.2.

3.10.2 Installed equipment and power tools operation. Using both external and shop generated power, operate all installed equipment and power tools.

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3.10.3 Roadability. When loaded with tools and equipment as specified in 3.2, 3.6.1 and 3.6.2, the shop shall withstand driving on unimproved roads, cross-country trails, and over a chatter course, and a diagonal ditch course. During or following driving, there shall be no damage to or displacement (see 6.8.2) of any component, accessory, part, or tool installed in or on the shop, and no evidence of defective welded, bolted, or riveted connections or accumulation of dust, mud or water in the interior of the shop.

3.11 Environmental conditions.

3.11.1 Heating system. Operate the heater (see 3.19) continuously for 15 minutes to check its performance.

3.11.2 Precipitation. The shop shall withstand accumulation of water to the interior of the shop or damage to components when exposed to natural or simulated rain fall, that which would inhibit shop operations.

3.11.3 Temperature (storage and operating).

3.11.3.1 Storage. The shop with all equipment shall be capable of withstanding an ambient temperature range of -50 ±5 degrees Fahrenheit (F) in storage without encountering any adverse effects or damage. The complete shop shall be operated when returned and stabilized to ambient temperature.

3.11.3.2 Operational temperature(s). The shop shall perform as specified below in any ambient temperature between -25 ±2 degrees F and +120 ±5 degrees F:

(a) Following a 24 hour soak period at -25 ±2 degrees F and while at that temperature operate the installed heater, all portable electric power tools, flood lights, trouble lights and electric welder from the shop installed power source. Slave starting of the government furnished vehicle may be attempted if starting difficulties are encountered (see 6.8.8). When returned to ambient temperature the complete shop shall be operated.
(b) Following a 24 hour soak period at +120 ±5 degrees F and while at that temperature, operate all shop equipment except the installed heater from the shop installed power source.

3.11.4 Sound level. The sound level of the shop shall be in accordance with MIL-STD-1474 Category D for steady-state noise when operating shop equipment. Hearing protectors need not be supplied with the shop equipment. Hazard signs, if required, shall state: "Caution: Hearing Protection Required" (see 6.13).

3.12 Reliability. The shop equipment shall demonstrate a mission reliability of at least 0.90 when measured with mean time between failures (MTBF).

3.13 Availability. The shop equipment availability shall be not less than 95 percent.

3.14 Maintainability. The shop equipment shall require a minimum of operational and in-storage maintenance. Replacement components shall be used to the maximum extent possible. Maintainability shall be measured in terms of maintenance ratio. Maintenance ratio shall be not greater than 0.12.

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3.15 Fuels, lubricants, and coolants. The shop shall be assembled and tested with the products of the type and grade specified in Table I for the applicable ambient air temperature. The shop shall operate as specified herein when furnished with these products (see 6.5).

TABLE I. Service Products.

Item	Application	Specification	Type and Grade
1	Engine:		
	(a) Heavy Duty.	MIL-L-2104	10, 30
	(b) Arctic.	MIL-L-46167	One Grade
2	Fuel:		
	Diesel Fuel	VV-F-800	
3	Grease:		
	(a) Automotive.	MIL-G-10924	One Grade
	(b) High Temperature.	MIL-G-23827	One Grade
4	Hydraulic-system Oil:		
	(a) Oil.	MIL-L-2104	10
	(b) Oil (low temperature).	MIL-L-46167	One Grade
5	Lubricating Oils:		
	Universal.	MIL-L-2105	80, 90, 140
6	Antifreeze:		
	(a) Ethylene Glycol.	O-A-548	One Grade
	(b) Arctic.	MIL-A-11755	One Grade
7	Brake Fluid.	VV-B-680	One Grade

3.15.1 Oil lubrication. Unless otherwise specified herein (see 6.2.1), all components shall be filled to the operating level with military oils designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

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3.16 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.17 Identification marking and instruction plates.

3.17.1 Electromagnetic interference (EMI). A permanent warning notice, indicating that EMI could result if electric power tools/equipment are operated in close vicinity to electronic communication or surveillance equipment, shall be included and displayed in a conspicuous place(s) on the shop van exterior entrance(s) and on the master power control panel.

3.17.2 Identification marking. The shop shall be identified in accordance with MIL-STD-130.

3.17.3 Instruction plates. Instruction plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material.

3.18 Treatment and painting. The portions of the shop normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type G, if nonferrous metal or Type F in ferrous metal, and drawing 11021251.

3.19 Government-furnished property. Unless otherwise specified (see 6.2.1), the Government shall furnish the property as specified in Table II (see 6.4).

TABLE II. Government-furnished property.

Item No.	Description	Identification	Quantity for each shop
1	Heater, space, multifuel with blower, 60,000 BTU/hr for vertical mounting and operation from 120-volt AC, 50/60 hertz, single-phase	NSN 4520-00-683-8595, MIL-H-11511, Type I	1

3.20 Workmanship.

3.20.1 Castings and forgings. All parts, components, and assemblies of the shop which include castings and forgings shall be clean of harmful and extraneous material such as sand, dirt, sprues, scale and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.20.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of the material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of metal may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated surfaces shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners except where sharp edges and corners are required or where they are not detrimental to safety.

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3.20.3 Welding. All weldments shall conform to 11021065 drawing, type and class as specified.

3.20.4 Riveting. All riveted connections shall conform to 11021066 drawing, Class III, Table III.

3.20.5 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching threaded area securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to asure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with referenced specifications, standards, and drawings as applicable.

4.1.3 Disassembly inspection. Any first article test shop failing a performance test shall be disassembled in the presence of a Government representative to the extent necessary to determine the cause of the failure. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

4.1.4 Parts and components. Parts and components shall be inspected as a minimum in accordance with the Quality Assurance Provisions (QAP) provided by the Government (see 6.8.1). The QAP's specify the characteristic(s) requiring inspection, the sampling plan and the basis for acceptance and rejection (see 6.7). For characteristics which are not listed in the QAP, and where a QAP does not exist, the contractor shall comply with the requirements on the product drawing.

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4.1.5 Preparation for examination and tests. Prior to examination or test, the shop shall be lubricated with oils and greases specified in 3.15. Oils shall be those designated for use in the ambient temperature at which the tests are being performed.

4.2 Classification of inspection. The inspection shall be classified as follows:

(a) First article inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Packaging inspection (see 4.8).

4.3 First article inspection.

4.3.1 Examination. The first article shall be examined for the defects marked "X" in Column 3 of Table III. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

TABLE III. Examination schedule.

Defect	Requirement Paragraph	First Article	Initial Production	Quality Conformance
1	2	3	4	5
Assembly incomplete or incorrect.	3.2	X	X	X
Coated cloth inclosure do not fit as specified when erected.	3.2.1	X	X	-
Material not as specified.	3.4	X	-	X
Dimensions, weight and lifting not as specified.	3.5	X	X	- -
Body and equipment not as specified.	3.6 thru 3.7, 3.19	X	X	X
Parts, components, tools or equipment missing or not functioning as specified.	3.8, 3.8.1	X	X	X
Stowage or tiedowns not as specified.	3.8	X	X	-
Safety devices and guards missing or not as specified.	3.9	X	X	X
Lubrication not as specified.	3.15, 3.15.1	X	X	X
Fungus and moisture resistance not as specified.	3.16	X	-	X

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TABLE III. Examination schedule (continued).

Defect	Requirement Paragraph	First Article	Initial Production	Quality Conformance
1	2	3	4	5
Identification marking, warning and instructions, notices and plates not as specified.	3.17	X	X	X
Treatment of painting not as specified.	3.18	X	X	X
Workmanship not as specified.	3.20	X	X	X

4.3.2 Tests. The first article shall be tested as specified in Column 4 of Table IV. Failure of any test shall be cause for performing the disassembly inspection as specified in 4.1.2.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production shops will be selected at random by the Government from shops being produced from production tooling and will be examined for the defects marked "X" in Column 4 of Table III and subjected to the tests marked "X" in Column 5 of Table IV to determine conformance to the requirements of this specification. The inspection shall be performed by the Government at a site selected by the Government. Acceptance of an initial production shop shall not exclude the remaining shops from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production shop to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production shop and shall be cause for refusal by the Government to continue acceptance of production shops until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on shops previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection shall be considered prima facie evidence that all shops accepted prior to the completion of initial production inspection are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.4.2 Weight. The complete shop shall be weighed using a scale. Weight shall be as specified in 3.5.2.

4.4.3 Lifting loading (wet weight). Using a scale, the lifting loading shall be as specified in 3.5.3.

4.5 Quality conformance inspection.

4.5.1 Examination. Each shop shall be examined as a minimum for the defects marked "X" in Column 5 of Table III. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each shop shall be subjected to the tests marked "X" in Column 6 of Table IV. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with Table III.

4.6.2 Tests. Tests shall be in accordance with Table IV. Sequence of testing shall be at the option of the test facility, except as specified in the paragraphs herein.

TABLE IV. Test schedule.

Examination or Test	Requirement Paragraph	Examination or Test Paragraph	First Article	Initial Production	Quality Conformance
1	2	3	4	5	6
Maximum power operation	3.10.1	4.6.2.1.1	X	X	X
Installed equipment and power tools operation	3.10.2	4.6.2.1.4	X	X	X
Overspeed	3.6.2.5	4.6.2.1.3	X	X	X
Road	3.10.3	4.6.2.1.4	X	X	-
Heating system	3.11.1	4.6.2.2.1	X	X	X
Rain	3.11.2	4.6.2.2.2	X	X	-
Temperature (storage and operation)	3.11.3	4.6.2.2.3	X	X	-
Sound level	3.11.4	4.6.2.2.4	X	X	-
Reliability	3.12	4.6.2.3	-	X	-
Availability	3.13	4.6.2.4	-	X	-
Maintainability	3.14	4.6.2.5	-	X	-

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4.6.2.1 Performance tests.

4.6.2.1.1 Maximum power operation. Load the unit to maximum rated load of 25 kw while operating at 208-volt, three phase, four wire, ac power. Apply a straight resistance ac load for two hours in ten minute cycles of continuous load for six minutes followed by four minutes at no-load. Nonconformance to 3.10.1 shall constitute failure of this test. The quality conformance test shall consist of continuous load for six minutes and may be performed as a part of 4.6.2.1.2.

4.6.2.1.2 Installed equipment and power tools operation. Using both external and shop generated power, operate all installed equipment and power tools for a period of at least 2 minutes (see 3.10.2). Failure of the power unit to compensate for high starting currents or inability of installed equipment or power tools to function shall constitute failure of this test.

4.6.2.1.3 Overspeed. Conduct this test at the conclusion of the test specified in 4.6.2.1.2. Pull the governor control to increase engine speed until the overspeed control stops the engine. Observe the AC frequency at which the overspeed control shuts-down the engine. Failure of the overspeed control to stop the engine when the frequency exceeds 66 Hz $\pm$1 Hz or inability to restart the engine following resetting of the control shall constitute failure of this test.

4.6.2.1.4 Road tests. Load the shop with the tools and equipment specified herein and perform the following tests, per requirements of 3.10.3.

(a) Unimproved roads. Drive the shop 350 miles on unimproved roads at maximum safe speeds. The course shall consist of poorly maintained gravel and dirt roads with rutting and occasional potholes.

(b) Cross-country trails. Drive the shop 150 miles on cross-country trails at maximum safe speeds. The course shall consist of native soil with rock added as required to maintain passability. Surface conditions shall range from smooth to rough with rutting, potholes and depressed areas.

(c) Chatter course. Drive the shop over a chatter course until at least 2,600 either upward or downward accelerations of magnitudes specified herein have been applied to a rear wheel center. The course shall be of construction that the acceleration magnitudes indicated below can be obtained by adjusting the vehicle speed without the frequency of the accelerations exceeding 15 cycles per second (cps) or becoming less than 3 cps. The maximum acceleration magnitude indicated below shall not be exceeded, and accelerations having magnitudes less than the minimum shall be disregarded. The course shall be of such length that a wheel center receives in one pass over the course at least 50 consecutive vertical acceleration cycles of any magnitude less than the maximum specified herein. Measure the accelerations by a trial test to establish the required vehicle test speed. Then disconnect the recording instruments and complete the test at established speed. The acceleration values used shall be those of the wheel center receiving the highest accelerations. Means for assuring that the vehicle test speed is held constant throughout the test shall be provided. Accelerations shall be measured on a running-gear member, usually an axle or spindle, that has a motion identical to that of the wheel center. The magnitude of the acceleration shall be not less than four times the acceleration of gravity (G) and not more than seven Gs.

(d) Diagonal ditch course. Drive the shop over a series of ditches until 600 reversal torsional frame stresses have been applied. The ditch profile shall be such as to allow vertical wheel travel of 18 inches from crest to root. The ditches shall be at 45 degrees to the centerline of the course and spaced a sufficient distance apart to allow only one rear wheel and one front wheel of opposite sides in the ditches at any one time. The shop shall be examined

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at the completion of each phase of the road test. Nonconformance to 3.10.3 shall constitute failure of this test.

4.6.2.2 Environmental.

4.6.2.2.1 Heating system test. Operate the heater continuously for not less than 15 minutes to check the operation of the heater, thermostat, and heat deflectors, (see 3.11.1). Any of the following that occurs as a result of faulty material supplied by the supplier or incorrect installation by the supplier shall constitute failure of this test:

(a) Malfunctioning of deflector or fuel supply line.
(b) Smoking, sooting, or back draft.
(c) Entrance of fuel vapors or exhaust gases into the interior of the shop.

4.6.2.2.2 Rain test. Park the shop on a horizontal surface and expose the shop to natural or simulated rain falling toward the shop at an angle of approximately 45 degrees to the horizontal. The volume of rain impinging on the surface shall be not less than two inches in a five minute period. Simulated rain shall originate from a source which produces droplets of uniform distribution and velocity similar to natural rain at the point of contact with the shop. Conduct the test as follows:

(a) Expose the top and left side to the rain as described above for a period of five minutes.
(b) Repeat step "a" with the top and front exposed.
(c) Repeat step "a" with the top and right side exposed.
(d) Repeat step "a" with the top and rear exposed.

Nonconformance to 3.11.2 shall constitute failure of this test. The canvas enclosure shall not be tested.

4.6.2.2.3 Temperature (storage and operation) tests. Place the shop in an ambient temperature of -50 ± 5 degrees F for a period not less than 72 hours (see 3.11.3.1). At the end of this period raise the temperature to -25 ± 2 degrees F for a period not less than 12 hours. At the end of this time period the temperature at -25 ± 2 degrees F, the following operational tests shall be performed (see 3.11.3.2):

(a) Inspect the entire unit for any damage which may have resulted from exposure to temperature extreme.
(b) Start and operate the alternator. It shall provide electrical power to both the 120/208 volt systems.
(c) Start and operate the heating system using diesel fuel Class DF-A having a cloud point not lower than -55 ±5 degrees F as specified in MIL-H-11511. The heater shall operate as required in 4.6.2.2.1.
(d) Start and operate individually all portable electric power tools, flood lights and trouble lights from the shop installed power source for a minimum of 2 minutes.
(e) Operate the electric welder in straight polarity and deposit two 6 inch bead at near minimum and at near maximum output. Repeat the test using reverse polarity.

After the conclusion of the operating test at -25 ±2 degrees F, stop all equipment, close up the unit, and raise the ambient temperature to +70 ±5 degrees F. When stabilized at +70 ±5 degrees F, start and operate individually all installed equipment and power tools from the shop installed power source for a minimum of 5 minutes.

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After the conclusion of the operating test at +70 ±5 degrees F, stop all equipment, close up the unit, and raise the ambient temperature to +120 ±5 degrees F for a period of 24 hours. After the 24 hours, soak at +120 ±5 degrees F, start and operate individually all installed equipment and power tools from the shop installed power source for a minimum of 2 minutes except the installed heater. Nonconformance to 3.11 shall be cause for rejection of the unit.

4.6.2.2.4 Sound level measurement. Operate the shop at governed speed with the alternator and air compressor operating. Measure the sound level at the tailgate work station while operating each of the power tools. Sound level in excess of Category D of MIL-STD-1474 shall require the installation of hazard signs as specified in 3.11.4.

4.6.2.3 Reliability. The initial production shop shall be tested for a total of ten missions to determine conformance to the requirements of 3.12. Data shall be gathered to evaluate the system reliability. Reliability is to be evaluated from MTBF and mission operating time (see 6.8.3 and 6.8.4).

$$\text{Reliability} = e^{-t/u}$$

where,

t = mission operating time, hour
u = mean time between failures (MTBF), hour
e = 2.718

The shop shall be considered to have demonstrated the required reliability if zero chargeable reliability failures occur (see 6.8.5).

4.6.2.3.1 Mission operation and test. A mission shall consist of the following:

(a) Drive the fully loaded shop set at maximum safe speeds for ten miles on unimproved roads and ten miles on cross-country trails.
(b) Using the instructions furnished with the shop, set-up the shop for operation to include rigging out of tent enclosure on the initial and final missions.
(c) Operate each installed equipment and power tool for a period not less than 5 minutes one time per mission, except as noted.
(d) Prepare the shop for movement to another location, to begin the next mission.

The following list of tools which are included in the shop shall be subjected to the 5 minutes test per mission.

(1) Air compressor.
(2) Drill, electric, portable, 1/4 in.
(3) Drill, electric, portable, 3/4 in.
(4) Drill, electric, portable, 1/2 in.
(5) Drilling machine.
(6) Floodlight, electric.
(7) Grinding kit, valve seat.
(8) Grinding machine, valve face.
(9) Grinding machine, utility.
(10) Honing machine, horizontal.
(11) Lathe, engine.
(12) Light, extensions, three conductor.
(13) Light, power tool.
(14) Milling-grinding-drilling-slotting attachment.
(15) Moisture stabilizer, welding electrode.
(16) Power, system, electrical.

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(17) Sander, disk, electric portable.
(18) Welder, operate as specified in 4.6.2.2.3(e).

Nonconformance with 3.12 shall constitute failure of this test.

4.6.2.4 Availability. During the performance of the Reliability Test as specified in 4.6.2.3, data shall be gathered to evaluate the system availability. Availability is to be evaluated with respect to operating time and corrective maintenance (see 6.8.4 and 6.8.6).

$$\text{Availability} = \frac{(\quad \text{MTBUMA} \quad)}{(\text{MTBUMA} + \text{MTTR})} \ (100 \text{ percent})$$

where,

MTBUMA = Mean time between unscheduled maintenance actions (unscheduled means corrective maintenance actions).

MTTR = Mean time to repair (use only times associated with corrective maintenance) (see 6.8.7).

The system availability requirement shall conform to 3.13. Nonconformance with 3.13 shall constitute failure to meet the availability requirement.

4.6.2.5 Maintainability. Maintenance operations required during first article and initial production tests shall be gathered to evaluate the system maintainability. Maintainability is to be evaluated with respect to total maintenance man-hours and total operating time during testing.

$$\text{Maintenance Ratio} = \frac{\text{TNMMT}}{\text{TOT}}$$

where,

TNMMT = Total number of man-hours of maintenance (scheduled maintenance and unscheduled repairs) during testing.

TOT = Total operating time (see 6.8.4).

Any support elements, i.e., technical publications, repair parts, special tools, etc., provided under the contract for assisting or conducting maintenance shall be evaluated. The system maintenance ratio shall conform to 3.14. Nonconformance with 3.14 shall constitute failure of this test.

4.7 Inspection comparison. The Goverment may select shops at any time during the contract production period and subject these shops to the examination specified in column 2 of table II and the tests specified in column 2 of table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Shops will be selected at random from those which have been accepted by the Government and will not include the previously inspected first article or initial production model. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison shop to meet any requirement specified herein during and as a result of the examination and test specified in 4.7 shall be cause for rejection of the inspection comparison shop and shall be cause for refusal by the Government to continue acceptance of production shops until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on shops previously produced and accepted under the contract. Any deficiencies found as a result of inspection comparison inspection will be considered prima facie evidence that all shops accepted after approval of the initial production shop are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Packaging inspection. The items shall be inspected before and after packaging to determine compliance with the requirements of Section 5 herein.

5 PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Industrial, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Methods. Methods of preservation indicated herein by symbols shall conform to the applicable symbol listed in MIL-P-116.

5.1.1.3 Sets or kits. When items are furnished in sets or kits with a carrying case, the items shall be preserved as specified herein and, when possible, replaced in the case.

5.1.1.4 Items having mounting brackets or holders. Items furnished with mounting brackets or holders shall be preserved as specified herein and, when possible, replaced in the bracket or holder and secured to prevent movement and damage.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces of the components, not specifically specified herein, requiring the application of a contact preservative in accordance with MIL-P-116 shall be preserved as follows.

5.1.1.5.1 Unfinished surfaces. Unfinished surfaces shall be coated with Type P-1 preservative except that cold finished and hot rolled steel shall be coated with Type P-7 preservative.

5.1.1.5.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with Type P-2 or P-6 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2 and secured in place with tape conforming to PPP-T-60, Type IV.

5.1.1.6 Semitrailer. The semitrailer with body shall be preserved in accordance with the Level A (mobile) preservation and packaging requirements of MIL-STD-281 with the following exceptions and additional requirements.

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5.1.1.6.1 Engine. The engine and engine accessories shall be preserved in accordance with the Level A requirements of MIL-E-10062, Type I, Method I.

5.1.1.6.2 Batteries. Batteries shall be secured in place in the battery carrier as specified in MIL-STD-281.

5.1.1.6.3 Cooling system. The cooling system shall be preserved by the water and antifreeze procedure as specified in MIL-STD-281.

5.1.1.7 Shop enclosure. Each section of the coated cloth shall be compactly and neatly folded, then wrapped with paper conforming to PPP-B-1055, Class C1 and the wrap secured with tape conforming to PPP-T-60, Type IV. The tent pins shall be packaged in a close-fitting box conforming to PPP-B-636. The coated cloth and tent pins shall be placed in the storage compartment provided. The unassembled framing shall be placed and secured on the roof as provided for.

5.1.1.8 Alternator. The alternator shall be preserved in accordance with MIL-E-16298, Level A, using the alternate method as specified for assembled machines.

5.1.1.9 Compressor. The air compressor and components shall be preserved and packaged in accordance with MIL-C-3600, Level A.

5.1.1.10 Handtools and tool sets. Handtools and tool sets shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.1.1.11 Belts and pulleys. All drive belts shall be released from tension or removed. Unpainted, unplated ferrous metal surfaces of the pulley shall be coated with primer conforming to TT-P-664. Removed belts shall be secured convenient to the pulley for which intended.

5.1.1.12 Space heater. The heater shall be completely drained of fuel.

5.1.1.13 Machine tools, attachments, and accessories. Machine tools such as the lathe, milling attachment, hydraulic press, bench grinder, grinding machine, drilling machine and other attachments, components, and accessories shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-M-18058. MIL-M-18058 shall also be used as a guide in the selection of preservatives and materials to be applied to items of similar construction not listed herein.

5.1.1.14 Chain hoist. The chain and any other unpainted metal surface of the hoist shall be coated with Type P-2 preservative. The hoist shall be wrapped in paper conforming to MIL-B-121, Grade A, Type I, Class 2, and packaged in a close fitting box conforming to PPP-B-636, Domestic Type.

5.1.1.15 Items furnished in commercial containers. Items such as pigment blue, sealing compound, welding electrodes and any other items commercially furnished in sealed can, pails, drums, jars, or tubes shall not require preservation.

5.1.1.16 Method III. The fitting kit, rubber hose, insert and assembly kit, electrical connector repair kit, temperature indicating compound, hardware items, and coupling halves shall be individually packaged Method III.

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5.1.1.17 Method IC-1. Technical publications and connector adapters shall be preserved Method IC-1.

5.1.1.18 Method IC-2. The first aid kit, flood light, hydraulic pump, spindle adapter, valve guide brush, and spark plug cleaner shall be individually preserved Method IC-2. The unit container shall be the contractor's commercially designed container or a close-fitting box conforming to PPP-B-566 or PPP-B-636, Class Domestic-variety, grade, and style optional.

5.1.1.19 Electrical wire. Electrical wire shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-C-12000.

5.1.1.20 Cable assemblies and electrical leads. Each cable assembly or test lead shall be coiled to not less than ten times its diameter and secured with cotton tape or twine.

5.1.1.21 Hydraulic jacks and lifts. The hydraulic-fluid supply tanks of hydraulic control systems shall be filled to the operating level with hydraulic fluid required by the technical manual supplied with the equipment. The systems shall be operated to insure complete coverage of the systems with the fluid. Unpainted exterior surfaces shall be coated with Type P-1 preservative.

5.1.1.22 Locks and keys. The interior of locks shall be coated with powdered graphite and the locks operated to assure penetration of the graphite. The locks used for securing the doors shall be protected against the elements by wrapping the locks in barrier material conforming to MIL-B-121, Grade A. The wrap shall be secured with tape conforming to PPP-T-60, Type IV. The keys shall be preserved Method III and forwarded to destination separately by registered mail. A cloth or paper shipping tag containing the shop equipment identification number and the trailer registration number shall be attached to the related padlock keys for future positive identification.

5.1.1.23 Fire extinguisher. Unpainted exterior metal surfaces of the fire extinguisher requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The extinguisher shall be placed in the mounting bracket and secured.

5.1.1.24 Electronic and electrical equipment. Electronic and electrical equipment and accessories, shall be preserved and packaged in accordance with MIL-E-17555, Level A.

5.1.1.25 Cork and paper gasket sheets. Sheets of like description and size shall be rolled together to the minimum safe diameter and each roll wrapped in barrier material conforming to PPP-B-1055, class C-1.

5.1.1.26 Welding and soldering equipment. Welding and soldering equipment shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-W-45562.

5.1.1.27 Other tools and equipment. Tool and equipment items which are not specifically covered by packaging requirements of this specification shall be preserved and unit packaged in accordance with the applicable commodity specification or MIL-T-45542.

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5.1.1.28 Miscellaneous supplies and equipment. Items, equipment, and components not specifically identified herein, shall be preserved as specified for supplies and equipment of similar design and construction.

5.1.1.29 Consolidation. Components that cannot be placed in the drawer, toolboxes, or compartments, or secured by the mounting brackets or holders provided, shall be consolidated together in close-fitting boxes conforming to PPP-B-636 Domestic Type, variety, grade and style optional. The boxes shall be of a size to fit into the available space within the body. The contents shall be blocked, braced, and cushioned as applicable to prevent movement and damage to the contents within the box.

5.1.2 Level B. Preservation shall be as specified for Level A with the following exceptions.

5.1.2.1 Semitrailer. The semitrailer with mounted body shall be preserved in accordance with the Level B (mobile) preservation and packaging requirements of MIL-STD-281.

5.1.2.2 Handtools and tool sets. Handtools and tool sets shall be preserved in accordance with the Level B requirements of PPP-P-40.

5.1.2.3 Compressor. The air compressor and components shall be preserved and packaged in accordance with MIL-C-3600, Level B.

5.1.2.4 Welding and soldering equipment. Welding and soldering equipment shall be preserved in accordance with Level B preservation and packaging requirements of MIL-W-45562.

5.1.3 Industrial. Industrial preservation of the semitrailer with mounted body and all shop equipment shall be in accordance with ASTM D3951-82 and as specified herein. Components shall be placed and secured in the compartment provided. Toolboxes, drawers, mounting brackets, and holders shall be utilized to the greatest extent possible.

5.2 Packing. The semitrailer mounted shop, with equipment enclosed, shall be prepared for shipment mobile in a manner to assure carrier acceptance and safe delivery to destination at lowest rating. Items and equipment, installed in the shop body, and any other movable equipment or items, shall be cushioned, blocked, braced, and anchored to prevent damage to the equipment or to the body. All doors and drawers shall be secured to prevent accidental opening and in a manner to prevent pilferage of contents. Padlock keys shall be identified with the vehicle registration number and shall be sent by registered mail to the recipient of the shop equipment (see 6.10).

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Industrial. Marking for industrial packaging shall be in accordance with ASTM D3951-82.

6 NOTES

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6.1 Intended use. The general-purpose-repair shop is intended for use by personnel engaged in the maintenance and repair of military mechanical equipment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of the first article (see 3.2).
(c) When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(d) When initial production inspection is required and number of shops to be furnished (see 3.3).
(e) Government-furnished property if other than that listed in 3.19.
(f) Level of preservation and packaging required (see 5.1).
(g) Gross weight if other than that listed (see 3.5.2).
(h) Components not filled with oil (see 3.15.1).

6.3 The contracting officer should arrange to furnish information concerning items specified in set SC 4940-95-CL-B02 (see 3.2).

6.4 Technical interface data. The contracting officer should arrange to provide information concerning technical interface for the government furnished semitrailer specified in 3.19.

6.5 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the shop covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91 (see 3.15).

6.6 Government-furnished property. The contracting officer should arrange to furnish the contractor the property specified in 3.19.

6.7 Quality assurance provisions (QAP). The contracting office should require the supplier to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum lot size, sample size, drawing requirement, number and type of deficiencies found, quantity approved, quantity rejected and corrective action taken when applicable."

6.8 Definitions.

6.8.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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6.8.2 Displacement. Displacement is defined as dislodgement of any tool or item of equipment from its designated position. Movement of a tool or item of equipment within its designated position without any dislodgement is permitted (see 3.8 and 3.10.3).

6.8.3 Mean-time-between-failure (MTBF). For a particular interval, the total functioning life of a population of an item divided by the total number of failures within the population during the measurement interval. The definition holds for time, cycle, miles, events, or other measure of life units (see 4.6.2.3).

6.8.4 Operating time (OT). Defined to be an 8-hour period per mission. This time does not include scheduled maintenance or unscheduled repair time. If the mission profile is completed in less than 8 hours, the balance is considered to be standby time (see 4.6.2.3 and 4.6.2.5).

6.8.5 Chargeable reliability failure. A chargeable reliability failure is defined as any malfunction which cannot be corrected by general purpose repair within five minutes by adjustment, repair, or replacement action using controls on the equipment, tools, or parts, and which causes or may cause:

(a) Failure to start operation.
(b) Cessation of operation.
(c) Degradation of performance below specified levels.
(d) Damage to the shop by continued operation.
(e) Safety hazards to personnel.

6.8.6 Corrective maintenance action. Any test incident that requires unscheduled maintenance time and includes both reliability chargeable failures and nonchargeable incidents (see 4.6.2.4).

6.8.7 Mean time to repair (MTTR). The total corrective maintenance time in clock hours divided by the total number of corrective maintenance actions (see 4.6.2.4).

6.8.8 Slave starting. External electrical power applied to a vehicle whose batteries have become discharged/depleted which may receive electrical battery voltage from an outside source through a slave cable plugged into the slave receptacle.

6.9 Environment testing. In lieu of requiring the contractor to perform the tests specified in 4.6.2.2.2 the contracting officer may accept certification that the equipment furnished with the shop will withstand storage in the temperatures specified, without malfunction, damage, or degradation of performance.

6.10 Packing. Levels of packing have not been included for semitrailer and mounted equipment, as the nature of the item is such that only mobile packing is appropriate. For purposes of preservation and packing level markings, the level of packing should be considered Level A.

6.11 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

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6.12 Provisions. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of shops.

6.13 Ear protection. When the noise level exceeds Category D of MIL-STD-1474, the operator's manual for the shop should include appropriate instructions for use of hearing protection when using equipment that produces noise above 85 decibel (dB)(A) inside and outside of the shop, and within a distance from the shop at which 85 dB(A) will be met.

6.14 Subject term (key word) listing.

Environmental conditions
Maintainability
Mission
Overspeed
Reliability
Repair shop
Roadability
Set components

6.15 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

Custodians:
Army - AL
Navy - YD

Review activity:
Navy - MC

Preparing activity:
Army - AL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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Check the source to verify that this is the current version before use.